NLP 245 Project Proposal Instructions*

Due: February 5 2026, 11:59 PM

1 Introduction

Each team will submit a project proposal worth 10% of your grade. The focus will be on LLM Agents. Please refer to the group formation assignment for some example topics. The best outcome of the project is that the final paper is publishable, reaching the level of short or workshop papers at major NLP conferences.

2 Required Group Meetings

To ensure everyone is on the right track, we are **requiring meetings** with the course staff. These will be held on:

- Thursday January 29th, 12:35 - 1:15 PM, Room 2235
- Thursday January 29th, 3:15 - 4:00 PM, Room 2235
- Wednesday February 4th, 3:30 - 5:00 PM, Room 3212

Each group must attend one of these sessions, with all group members present. We ask that you bring ideas for topics to the meetings so we can discuss and narrow them down to something feasible. The more preparation you do, the better the proposal you will have, and the more potential there is to produce something novel that excites the research community.

3 Proposal Information

Please include the following information at the top of your PDF proposal:

1. Title: The title of the project (which can be changed later)
2. Team members: List all of your team members' names and @ucsc.edu email addresses.

3.1 Paper Summary (1-2 pages)

Paper selection and eligibility There are several approaches to picking a final project topic. For example, you may take an existing problem from class and generate fresh ideas. We encourage you to carefully check the published research papers and consider whether there is any challenge left from the publications or if you can improve the provided strategy. As a result, the first component of your proposal should be a synopsis of a research publication related to your topic. If you want to apply a specific method to a new problem, you may read the paper that explains the process. You might select the publication that offers the original agent if you are constructing a new variation of a given agent. There are additional possibilities; use your best judgment to select what appears to be the most relevant paper. Any high-quality research article is acceptable in general. We recommend looking into a peer-reviewed forum (i.e., conference or journal) or for high-quality papers that are not yet published (e.g., on arXiv). Asking the following questions can help choose a paper and a project topic:

- What is the problem? Why is this problem exciting and important?
- Is there an existing approach? How is your idea different from others? Why is your idea challenging?
- How to evaluate your idea? What data can be used for evaluating the proposed approach? Is the data set available?

*Some content is from Stanford CS224n.

- What software and resources are available to help you implement your idea? Does the paper provide open-source datasets and code? You should have access to the datasets and the computation resources to run any necessary experiments for your project (e.g., UCSC NLP Cluster, Google Colab, and Google Cloud Credits).

We know this is very open-ended, but this is the nature of research. We are requiring group meetings so that you will have guidance and be able to shape your ideas into executable papers. Please discuss potential topics with your group and bring many options and questions so we can help guide you and make something great together!

Summary Write a two-page summary of the paper that a NLP 245 student could understand. You should explain any new terminology introduced outside the course and avoid describing course topics. You may add a diagram if you consider it to be necessary, but it should not take up more than one-third of a page. Prioritize communicating the most significant material and concepts from the article in your 2-page summary. In particular, when evaluating your summary, we will explicitly look for a discussion of each of the following aspects:

- Bibliographical info: Title, authors, publication venue (or preprint server, like arXiv), publication year, and URL. Put this information at the top of your summary
- Background
- Summary of contributions
- Limitations and discussion
- Why this paper?
- Wider research context

This summary should not exceed two pages. Use your discretion to assess what material from the article is essential when covering each of the parts listed above. The proposal template contains additional information.

3.2 Project Description (1-2 pages)

In this section, you will outline your project's objectives. It's acceptable if your goals change, but your proposal should provide a starting strategy. It would help if you addressed the following questions in this section.

- Describe the primary goal(s) of the project.
- What task(s) will you tackle?
- What data will you use?
- What method(s) will you use?
- What baselines will your method(s) be compared to?
- How will you evaluate your results?

Additional details are provided in the proposal template. Additionally, you might use this opportunity to write the introduction and related work portion of your final paper.

4 Submission

Please only submit one proposal per group via Canvas. The deadline to submit your proposal is **February 5, 2026 11:59 PM**. Be sure to use the provided proposal template.

5 Other Information and Tips

Grading When grading your project proposals, we are looking for clarity, depth, and well-considered analysis. We also want to provide input; the more effort and thought you put into your proposal, the more quality comments we can provide to assist you in improving your project. We'll decide whether your project meets the proposal requirements and may advise changes after reading it (e.g., not practical given the schedule).

Contribution Every team member should make a significant contribution. In the final paper, we ask for a statement of what each team member contributed to the project. Team members will typically get the same grade, but we may differentiate in extreme cases of unequal contribution. You can contact us in confidence via EdStem in the event of unequal contribution.

External resources Describe and cite existing code libraries or packages you utilized in your project, and specify which portions of the project you contributed to and which aspects were implemented by others. If you have any questions regarding how to acknowledge the efforts of others properly, please get in touch with the course staff, and we will gladly assist you. Of course, you must not violate any laws or breach any terms of service.